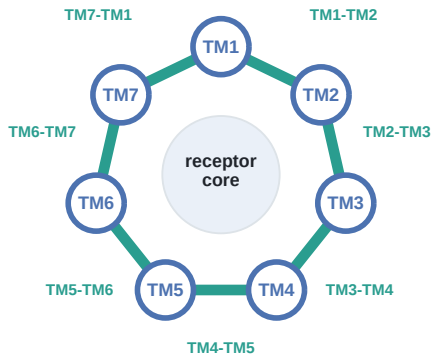


A Seven TM-helix gateways

Top view; TM assignments from GPCRdb



One record per neighboring helix pair

7 interfaces x 4 reported metrics

Identity contract

Same interface keys in every included system

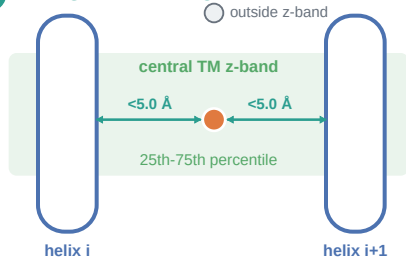
Missing record: null + reason code

Never substitute a numerical zero

B Frame-level classification

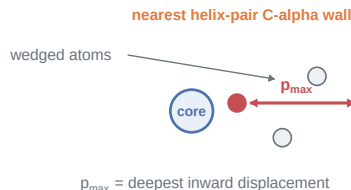
Two decisions are applied to each sampled frame

WEDGED LIPID ATOM



distance to BOTH helices $< 5.0 \text{ \AA}$
AND atom z lies inside the central band

OPEN FRAME



$p_{\max}$ = deepest inward displacement

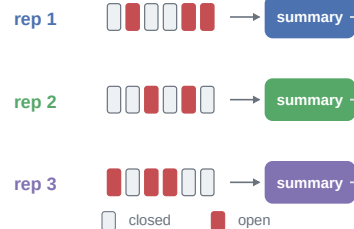
$p_{\max} \geq 0.5 \text{ \AA}$
frame classified as OPEN

C Replica aggregation

Frames $\rightarrow$ 3 replicas $\rightarrow$ system record

sampled frames

4 metrics



SYSTEM-LEVEL GATEWAY RECORD

mean of the 3 replica summaries
for each interface and metric

95% bootstrap interval
1,000 resamples; fixed seed

penetration

mean $p_{\max}$ ($\text{\AA}$)

penetration_p90

90th percentile ($\text{\AA}$)

open_fraction

open frames / all frames

occupancy

wedged atoms / frame